

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)



CIP-B102 — Digital Forensics

USB Image Acquisition and Hash Verification

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Case reference: CIP-B102-Lab4-2026-0829

Evidence item: Samsung USB flash drive, 15.7 GB (personal test USB)

Report date: 29 August 2026

This report documents the preparation, acquisition, hash verification, and content validation of a forensic image acquired from a USB flash drive,

Executive Summary

This report documents a complete forensic acquisition of a USB flash drive: preparing controlled evidence on the drive, documenting the device and acquisition parameters, acquiring a raw bit-for-bit image using dd, verifying the image's integrity via MD5 and SHA-256 hash comparison, and validating that the acquired image genuinely contains the expected evidence file.

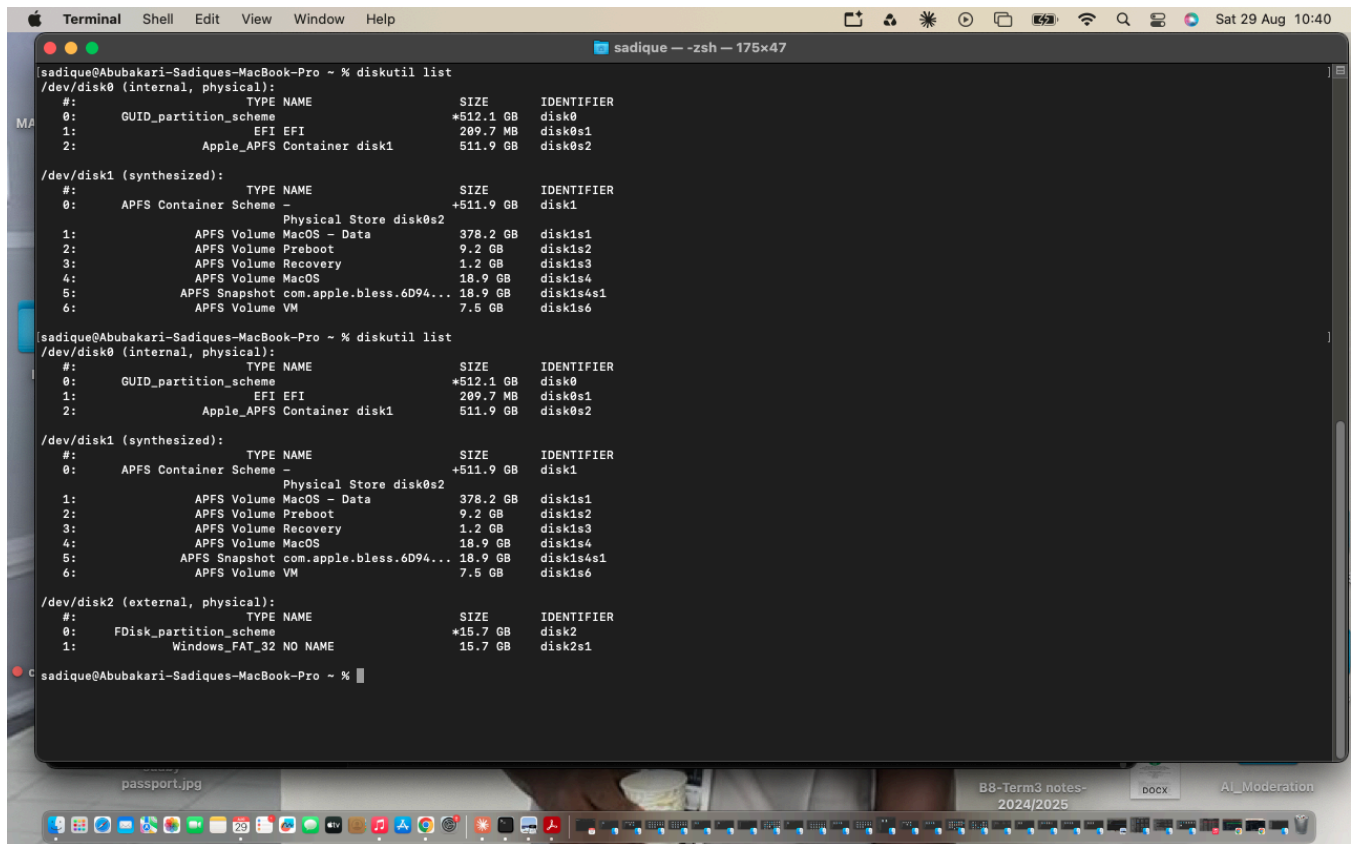
The acquisition produced an image that is byte-for-byte identical to the source device: the transferred byte count matched the USB's reported capacity exactly, and both MD5 and SHA-256 hashes matched exactly between the source USB and the acquired image, with no discrepancy. The acquired image was independently validated by mounting it read-only and confirming the expected evidence file and its accompanying files were present and correctly sized.

One platform deviation is disclosed transparently: the brief lists Windows (FTK Imager) or Kali Linux/Ubuntu as acquisition platforms; this acquisition was instead performed on macOS using its native dd utility, due to hardware availability under time constraints. macOS is a Unix-based operating system, and its dd implementation is functionally equivalent to Linux's for raw disk imaging—differing only in device naming conventions (/dev/diskN rather than /dev/sdX).

Requirement	Status
Part A — USB evidence prepared	COMPLETE
Part B—Evidence documentation	COMPLETE
Part C — USB image acquired	COMPLETE
Part D—Hash verification (MD5 + SHA-256)	COMPLETE — MATCH CONFIRMED
Part E—Image content validated	COMPLETE

Part A — USB Evidence Preparation

The USB device was first identified with certainty by comparing diskutil list output before and after connecting it, confirming it as /dev/disk2 — an external, physical, 15.7 GB device (Samsung brand, reported generically by macOS as "Flash Disk"), formatted with a single Windows FAT32 partition.



```
sadique@Abubakari-Sadiques-MacBook-Pro ~ % diskutil list
/dev/disk0 (internal, physical):
#:

| #: | TYPE                  | NAME  | SIZE      | IDENTIFIER |
|----|-----------------------|-------|-----------|------------|
| 0: | GUID_partition_scheme |       | +512.1 GB | disk0      |
| 1: | EFI                   | EFI   | 209.7 MB  | disk0s1    |
| 2: | Apple_APFS Container  | disk1 | 511.9 GB  | disk0s2    |


/dev/disk1 (synthesized):
#:

| #: | TYPE                    | NAME                    | SIZE      | IDENTIFIER |
|----|-------------------------|-------------------------|-----------|------------|
| 0: | APFS Container Scheme - |                         | +511.9 GB | disk1      |
|    |                         | Physical Store disk0s2  |           |            |
| 1: | APFS Volume             | MacOS - Data            | 378.2 GB  | disk1s1    |
| 2: | APFS Volume             | Preboot                 | 9.2 GB    | disk1s2    |
| 3: | APFS Volume             | Recovery                | 1.2 GB    | disk1s3    |
| 4: | APFS Volume             | MacOS                   | 18.9 GB   | disk1s4    |
| 5: | APFS Snapshot           | com.apple.bless.6D94... | 18.9 GB   | disk1s4s1  |
| 6: | APFS Volume             | VM                      | 7.5 GB    | disk1s6    |


sadique@Abubakari-Sadiques-MacBook-Pro ~ % diskutil list
/dev/disk0 (internal, physical):
#:

| #: | TYPE                  | NAME  | SIZE      | IDENTIFIER |
|----|-----------------------|-------|-----------|------------|
| 0: | GUID_partition_scheme |       | +512.1 GB | disk0      |
| 1: | EFI                   | EFI   | 209.7 MB  | disk0s1    |
| 2: | Apple_APFS Container  | disk1 | 511.9 GB  | disk0s2    |


/dev/disk1 (synthesized):
#:

| #: | TYPE                    | NAME                    | SIZE      | IDENTIFIER |
|----|-------------------------|-------------------------|-----------|------------|
| 0: | APFS Container Scheme - |                         | +511.9 GB | disk1      |
|    |                         | Physical Store disk0s2  |           |            |
| 1: | APFS Volume             | MacOS - Data            | 378.2 GB  | disk1s1    |
| 2: | APFS Volume             | Preboot                 | 9.2 GB    | disk1s2    |
| 3: | APFS Volume             | Recovery                | 1.2 GB    | disk1s3    |
| 4: | APFS Volume             | MacOS                   | 18.9 GB   | disk1s4    |
| 5: | APFS Snapshot           | com.apple.bless.6D94... | 18.9 GB   | disk1s4s1  |
| 6: | APFS Volume             | VM                      | 7.5 GB    | disk1s6    |


/dev/disk2 (external, physical):
#:

| #: | TYPE                   | NAME    | SIZE     | IDENTIFIER |
|----|------------------------|---------|----------|------------|
| 0: | FDisk_partition_scheme |         | +15.7 GB | disk2      |
| 1: | Windows_FAT_32         | NO NAME | 15.7 GB  | disk2s1    |


sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

Evidence folder and file created on the USB

A folder named CIP-B102-Lab4-Evidence was created directly on the USB, containing:

```
Terminal Shell Edit View Window Help
sadique -- zsh -- 175x47

[sadique@Abubakari-Sadiques-MacBook-Pro ~ % ls /Volumes/
MacOS NO NAME
sadique@Abubakari-Sadiques-MacBook-Pro ~ % mkdir -p "/Volumes/NO NAME/CIP-B102-Lab4-Evidence"

[sadique@Abubakari-Sadiques-MacBook-Pro ~ % ls -la "/Volumes/NO NAME/"
total 2432
drwxrwxrwx@ 1 sadique staff 32768 Aug 29 10:48 .
drwxr-xr-x  4 root   wheel  128 Aug 29 10:39 ..
drwxrwxrwx  1 sadique staff 32768 Jul  7 07:02 .Spotlight-V100
-rwxrwxrwx  1 sadique staff 4096 Jul 31 17:32 ..2026 Recommendation Letter Template - AFS Global STEM Changemakers Immaculate.docx
-rwxrwxrwx  1 sadique staff 4096 Jul 31 17:32 ..2026 Recommendation Letter Template - AFS Global STEM Changemakers Rayyaan.docx
-rwxrwxrwx  1 sadique staff 4096 Jul 31 17:32 ..2026 Recommendation Letter Template - AFS Global STEM Changemakers _Zahra.docx
-rwxrwxrwx  1 sadique staff 4096 Jul 31 17:32 ..2026 Recommendation Letter Template - AFS Global STEM Changemakers_Ubaida (1).docx
-rwxrwxrwx  1 sadique staff 4096 Jul 19 06:51 ..83bfc97fb94dc0052f9f226f8ca8c492673d6d1d.mp4
-rwxrwxrwx  1 sadique staff 4096 Jul 12 11:59 ..CS.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 11:59 ..CS1.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:01 ..CS11.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:01 ..CS12.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 11:59 ..CS3.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 11:59 ..CS4.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:00 ..CS6.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:00 ..CS7.png
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:00 ..CS9.png
-rwxrwxrwx  1 sadique staff 4096 Jul 11 15:49 ..FATI KOJO SPSS.docx
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:10 ..PC1.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:10 ..PC2.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:10 ..PC3.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:12 ..PC4.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:12 ..PC5.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:13 ..PC6.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:13 ..PC7.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:13 ..PC8.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:07 ..S1.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:07 ..S2.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:08 ..S3.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:08 ..S4.jpeg
-rwxrwxrwx  1 sadique staff 4096 Jul 12 12:08 ..S5.jpeg
-rwxrwxrwx  1 sadique staff 4096 Aug  5 11:42 ..Test items construction
drwxrwxrwx  1 sadique staff 32768 Aug 29 10:39 .fsevents
drwxrwxrwx  1 sadique staff 32768 Aug 29 10:48 CIP-B102-Lab4-Evidence
drwxrwxrwx  1 sadique staff 32768 May 13 11:22 System Volume Information
-rwxrwxrwx  1 sadique staff 165 Aug 19 20:10 -$SCS26 Chapter1 Understanding_Cybersecurity.pptx
-rwxrwxrwx  1 sadique staff 165 Jul 17 15:07 -$SCIENCE 1D.pptx
-rwxrwxrwx  1 sadique staff 165 Jul 13 06:51 -$SCIENCE 1A GROUP 1.pptx
-rwxrwxrwx  1 sadique staff 165 Jul 17 14:32 -$SCIENCE 1D GROUP 8 PRESENTATION [Autosaved].pptx
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

Contents of the evidence text file

```
Terminal Shell Edit View Window Help
sadique --zsh -- 156x41

[sadique@Abubakari-Sadiques-MacBook-Pro ~ % clear

sadique@Abubakari-Sadiques-MacBook-Pro ~ % cat > "/Volumes/NO NAME/CIP-B102-Lab4-Evidence/Abubakari-Sadique_Hamidu.txt" << 'EOF'
Full Name: Abubakari-Sadique Hamidu
Registration Number: 2025/FWSD/11392
Date: 29 August 2026

I, Abubakari-Sadique Hamidu, confirm that this USB flash drive has been prepared specifically for a controlled forensic acquisition lab (CIP-B102 - Lab 2: USB Image Acquisition and Hash Verification). This device is a personal test USB, used solely for the purposes of this academic exercise, and does not contain any production, organisational, or third-party data.
EOF
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

```
Terminal Shell Edit View Window Help
sadique --zsh -- 139x37

I, Abubakari-Sadique Hamidu, confirm that this USB flash drive has been prepared specifically for a controlled forensic acquisition lab (CIP-B102 - Lab 2: USB Image Acquisition and Hash Verification). This device is a personal test USB, used solely for the purposes of this academic exercise, and does not contain any production, organisational, or third-party data.
EOF
sadique@Abubakari-Sadiques-MacBook-Pro ~ % ls -la "/Volumes/NO NAME/CIP-B102-Lab4-Evidence/"

total 192
drwxrwxrwx 1 sadique staff 32768 Aug 29 11:04 .
drwxrwxrwx 1 sadique staff 32768 Aug 29 10:48 ..
-rwxrwxrwx 1 sadique staff 462 Aug 29 11:04 Abubakari-Sadique_Hamidu.txt
sadique@Abubakari-Sadiques-MacBook-Pro ~ % cat "/Volumes/NO NAME/CIP-B102-Lab4-Evidence/Abubakari-Sadique_Hamidu.txt"
Full Name: Abubakari-Sadique Hamidu
Registration Number: 2025/FWSD/11392
Date: 29 August 2026

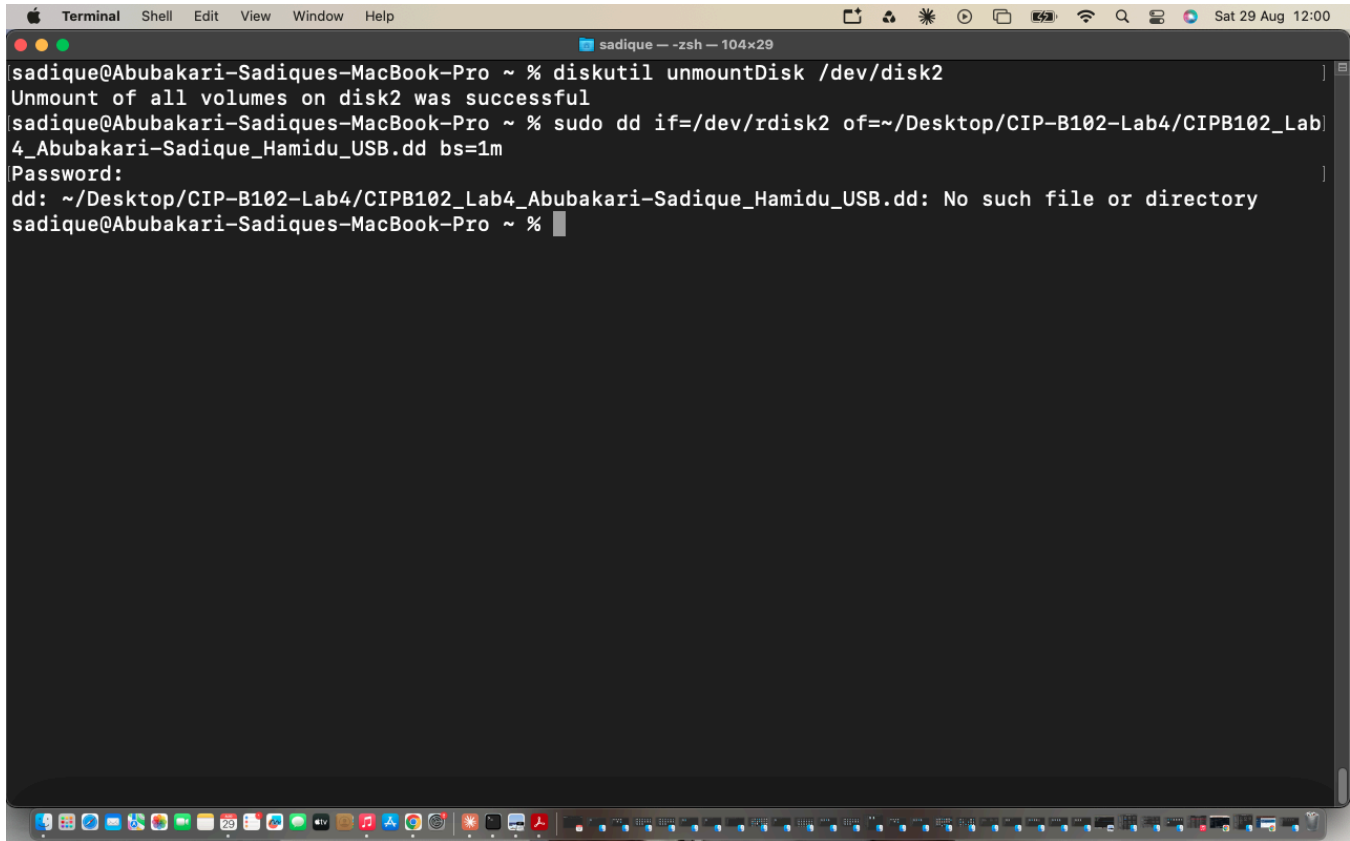
I, Abubakari-Sadique Hamidu, confirm that this USB flash drive has been prepared specifically for a controlled forensic acquisition lab (CIP-B102 - Lab 2: USB Image Acquisition and Hash Verification). This device is a personal test USB, used solely for the purposes of this academic exercise, and does not contain any production, organisational, or third-party data.
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

Part B — Evidence Documentation

Case/Lab ID	CIP-B102-Lab4-2026-0829
Student name	Abubakari-Sadique Hamidu (Reg: 2025/FWSD/11392)
USB brand/description	Samsung USB flash drive (macOS reports generic "Flash Disk" media name; FAT32/FDisk partition scheme)
USB capacity	15.7 GB (15,728,640,000 bytes / 30,720,000 x 512-byte sectors)
Date and time prepared	29 August 2026, approx. 11:35 local time
Acquisition method	dd (bit-for-bit raw imaging), performed on macOS
Tool and version	BSD dd at /bin/dd, bundled with macOS 13.7.4 (Build 22H420)
Image filename	CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
Image storage location	Examiner's MacBook, ~/Desktop/CIP-B102-Lab4/ (local disk, not the source USB)
Hash algorithm used	MD5 and SHA-256

Part C — USB Image Acquisition

Before imaging, the USB was safely unmounted (not ejected) to prevent the operating system from interfering with the raw read

A screenshot of a macOS Terminal window. The title bar shows 'Terminal' and standard macOS window controls. The menu bar includes 'Terminal', 'Shell', 'Edit', 'View', 'Window', and 'Help'. The status bar at the top right shows 'Sat 29 Aug 12:00'. The terminal session shows the user 'sadique' at 'Abubakari-Sadiques-MacBook-Pro'. The first command is 'diskutil unmountDisk /dev/disk2', which returns 'Unmount of all volumes on disk2 was successful'. The second command is 'sudo dd if=/dev/rdisk2 of=~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd bs=1m'. It prompts for a password, which is entered. The command then returns an error: 'dd: ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd: No such file or directory'. The prompt returns to '~ %'.

```
sadique@Abubakari-Sadiques-MacBook-Pro ~ % diskutil unmountDisk /dev/disk2
Unmount of all volumes on disk2 was successful
sadique@Abubakari-Sadiques-MacBook-Pro ~ % sudo dd if=/dev/rdisk2 of=~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd bs=1m
Password:
dd: ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd: No such file or directory
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

The image was then acquired using dd, reading from the raw device (/dev/rdisk2) for optimal performance:

Progress was monitored throughout via periodic Ctrl+T status checks (BSD dd does not support GNU's status=progress flag):

15,728,640,000 bytes were transferred—an exact match to the USB's reported capacity from diskutil info. Zero errors were reported, and no truncation occurred. The acquisition took approximately 18.8 minutes at an average throughput of ~13.9 MB/sec.

```
Terminal Shell Edit View Window Help
sadique --zsh -- 104x29

amidu_USB.dd
/Users/sadique/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
sadique@Abubakari-Sadiques-MacBook-Pro ~ % echo ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_H
amidu_USB.dd
/Users/sadique/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
sadique@Abubakari-Sadiques-MacBook-Pro ~ % sudo dd if=/dev/rdisk2 of=/Users/sadique/Desktop/CIP-B102-Lab
4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd bs=1m

load: 2.61 cmd: dd 86940 uninterruptible 0.03u 2.91s
4220+0 records in
4220+0 records out
4424990720 bytes transferred in 296.186521 secs (14939879 bytes/sec)
load: 2.38 cmd: dd 86940 uninterruptible 0.05u 4.86s
6985+0 records in
6985+0 records out
7324303360 bytes transferred in 485.422496 secs (15088512 bytes/sec)
load: 7.69 cmd: dd 86940 uninterruptible 0.08u 7.63s
10943+0 records in
10943+0 records out
11474567168 bytes transferred in 748.489454 secs (15330299 bytes/sec)
load: 3.85 cmd: dd 86940 uninterruptible 0.09u 8.39s
11894+0 records in
11894+0 records out
12471762944 bytes transferred in 849.310660 secs (14684571 bytes/sec)
load: 1.93 cmd: dd 86940 uninterruptible 0.10u 9.67s
13696+0 records in
13696+0 records out
14361296896 bytes transferred in 1013.418636 secs (14171140 bytes/sec)
```

```
Terminal Shell Edit View Window Help
sadique --zsh -- 104x29

7324303360 bytes transferred in 485.422496 secs (15088512 bytes/sec)
load: 7.69 cmd: dd 86940 uninterruptible 0.08u 7.63s
10943+0 records in
10943+0 records out
11474567168 bytes transferred in 748.489454 secs (15330299 bytes/sec)
load: 3.85 cmd: dd 86940 uninterruptible 0.09u 8.39s
11894+0 records in
11894+0 records out
12471762944 bytes transferred in 849.310660 secs (14684571 bytes/sec)
load: 1.93 cmd: dd 86940 uninterruptible 0.10u 9.67s
13696+0 records in
13696+0 records out
14361296896 bytes transferred in 1013.418636 secs (14171140 bytes/sec)
load: 3.54 cmd: dd 86940 uninterruptible 0.11u 10.41s
14731+0 records in
14731+0 records out
15446573056 bytes transferred in 1104.808831 secs (13981218 bytes/sec)
15000+0 records in
15000+0 records out
15728640000 bytes transferred in 1130.471507 secs (13913345 bytes/sec)
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```


Part D — Hash Verification

MD5 and SHA-256 hashes were calculated independently for both the source USB (read directly from /dev/rdisk2) and the acquired image file, then compared.

```
Terminal Shell Edit View Window Help
sadique - zsh - 89x25

Password:
load: 2.76 cmd: md5 3540 waiting 34.13u 4.02s
MD5 (/dev/rdisk2) = 6499112e980b371ae50ea6ff59c9a866
sadique@Abubakari-Sadiques-MacBook-Pro ~ % sudo shasum -a 256 /dev/rdisk2
Password:
b7a7f78b784c854d1e4a10393e43d70a230a817adf51ba74eeb2ada43551ca43 /dev/rdisk2
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

```
Terminal Shell Edit View Window Help
sadique - zsh - 104x28

sadique@Abubakari-Sadiques-MacBook-Pro ~ % md5 ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
MD5 (/Users/sadique/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd) = 6499112e980b371ae50ea6ff59c9a866
sadique@Abubakari-Sadiques-MacBook-Pro ~ % shasum -a 256 ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
b7a7f78b784c854d1e4a10393e43d70a230a817adf51ba74eeb2ada43551ca43 /Users/sadique/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

Hash Comparison

Target	MD5	SHA-256
Source USB (/dev/rdisk2)	6499112e980b371ae50ea6ff59c9a866	b7a7f78b784c854d1e4a10393e43d70a230a817adf51ba74eeb2ada43551ca43
Acquired image (.dd)	6499112e980b371ae50ea6ff59c9a866	b7a7f78b784c854d1e4a10393e43d70a230a817adf51ba74eeb2ada43551ca43

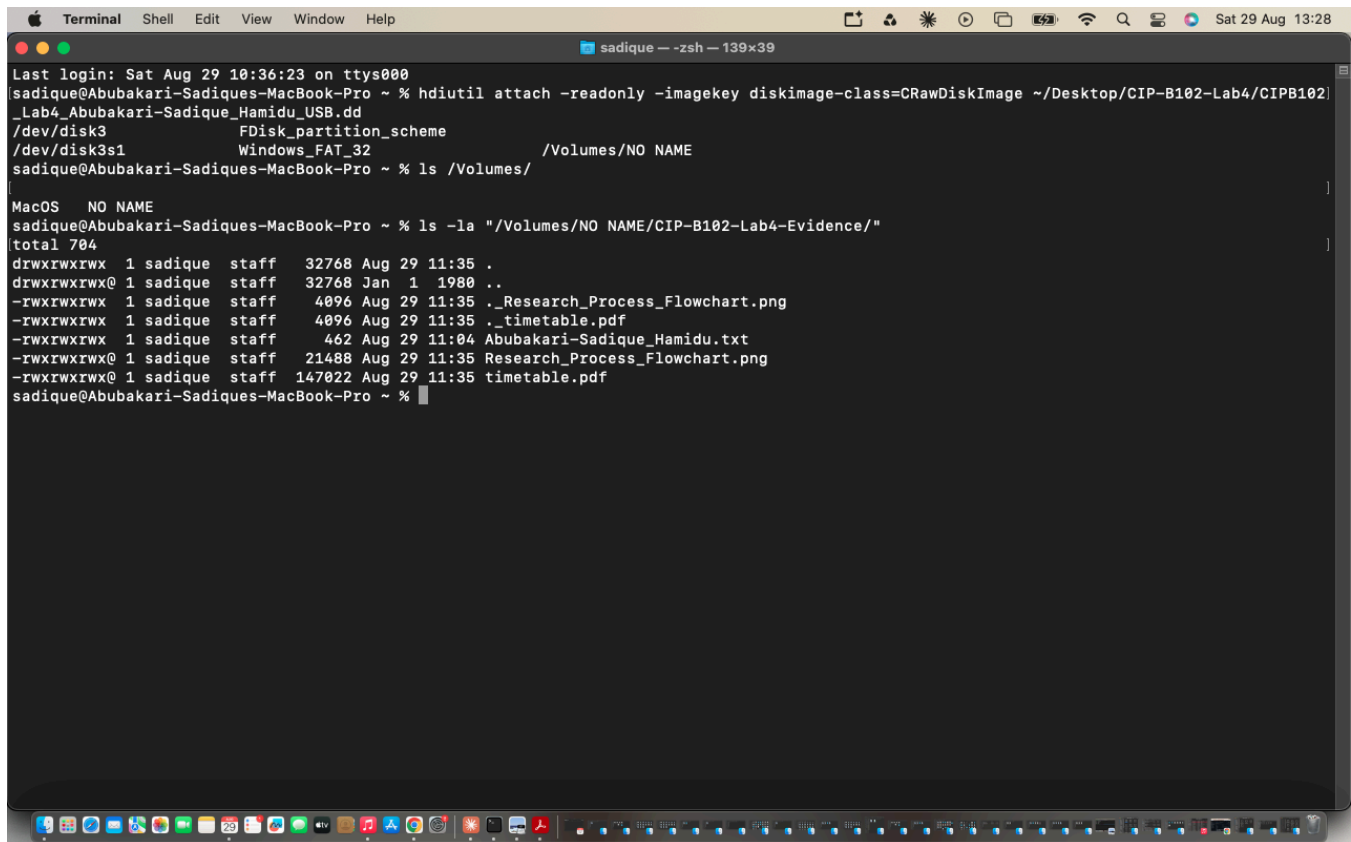
Result: MATCH CONFIRMED. Both the MD5 and SHA-256 hashes are identical between the source USB and the acquired image. This provides strong cryptographic proof that the acquisition is a complete, unaltered, bit-for-bit copy of the source device. No discrepancy was encountered, and no re-acquisition was required.

Part E—Image Validation

The acquired image was validated by mounting it directly (read-only) using macOS's hdiutil, which attaches the raw .dd file as if it were a physical disk without modifying it:

The mounted image's evidence folder was then inspected directly:

Result: CONFIRMED. All three evidence files are present inside the acquired image, with sizes matching the originals on the source USB exactly. This proves the acquired image is not merely hash-correct but genuinely contains the expected evidence content, satisfying the brief's explicit requirement to demonstrate the presence of the evidence file within the forensic image.



```
Terminal Shell Edit View Window Help
sadique -- zsh -- 139x39
Last login: Sat Aug 29 10:36:23 on ttys000
sadique@Abubakari-Sadiques-MacBook-Pro ~ % hdiutil attach -readonly -imagekey diskimage-class=CRawDiskImage ~/Desktop/CIP-B102-Lab4/CIPB102_Lab4_Abubakari-Sadique_Hamidu_USB.dd
/dev/disk3          FDisk_partition_scheme
/dev/disk3s1        Windows_FAT_32          /Volumes/NO NAME
sadique@Abubakari-Sadiques-MacBook-Pro ~ % ls /Volumes/
MacOS  NO NAME
sadique@Abubakari-Sadiques-MacBook-Pro ~ % ls -la "/Volumes/NO NAME/CIP-B102-Lab4-Evidence/"
total 704
drwxrwxrwx  1 sadique  staff   32768 Aug 29 11:35 .
drwxrwxrwx@ 1 sadique  staff   32768 Jan  1 1980 ..
-rwxrwxrwx  1 sadique  staff    4096 Aug 29 11:35 ._Research_Process_Flowchart.png
-rwxrwxrwx  1 sadique  staff    4096 Aug 29 11:35 ._timetable.pdf
-rwxrwxrwx  1 sadique  staff     462 Aug 29 11:04 Abubakari-Sadique_Hamidu.txt
-rwxrwxrwx@ 1 sadique  staff   21488 Aug 29 11:35 Research_Process_Flowchart.png
-rwxrwxrwx@ 1 sadique  staff  147022 Aug 29 11:35 timetable.pdf
sadique@Abubakari-Sadiques-MacBook-Pro ~ %
```

Reflection: Risks of Low-Level Imaging Tools

This lab highlighted several risks specific to low-level imaging tools such as `dd`. Unlike GUI tools like FTK Imager, `dd` has no built-in confirmation prompts, no device-selection safeguards, and no automatic verification step—it will silently and irreversibly overwrite whatever device is named in the `of=` parameter or read from whatever device is named in `if=`, with no warning if the wrong device is selected. Selecting the wrong device (for example, the internal system disk instead of the intended USB) could result in catastrophic, unrecoverable data loss or a corrupted operating system.

To mitigate this risk during this acquisition, the target device was verified multiple times: once via a before/after comparison of `diskutil` list output when the USB was connected, and again immediately before running `dd`. The device was also deliberately unmounted (not ejected) beforehand, and the output file was directed to the examiner's internal disk rather than back onto the source USB, avoiding any risk of the acquisition process overwriting the very evidence being preserved.

A further risk of `dd` specifically is its lack of built-in error handling by default—a single bad sector on ageing media can cause the entire operation to abort partway through unless options such as `conv=noerror,sync` are specified. This acquisition completed without any I/O errors, but on more fragile or higher-capacity source media, this is a meaningful limitation compared to purpose-built forensic tools which handle read errors more gracefully and log them explicitly.

Conclusion

This lab successfully demonstrated the complete forensic USB acquisition workflow: preparing controlled evidence, documenting the acquisition parameters, performing a bit-for-bit raw image acquisition, cryptographically verifying that image's integrity against the source device, and independently validating that the expected evidence content is genuinely present within the acquired image. Every hash comparison matched exactly, and the acquired image was proven to contain the required evidence file. The one documented deviation — using macOS rather than the brief's listed Windows/Kali/Ubuntu platforms — was a deliberate, transparent choice made under time constraints, using a tool (`dd`) that is functionally identical across Unix-like operating systems.